

Market Analysis Project Report

A data-driven investigation of order patterns, product performance, and reorder behavior to optimize marketing strategy and improve customer satisfaction

Executive Summary

This report presents a comprehensive analysis of online grocery ordering behavior for project CDACL-006, covering 63,100 customers, 49,355 products, 134 aisles, and 21 departments. The analysis was conducted across three integrated stages: data validation and exploration in MySQL, in-depth statistical and behavioral analysis in Python, and executive-level visual reporting in Power BI. The objective was not simply to describe historical order data, but to translate it into decisions that Marketing, Merchandising, and Operations teams can act on immediately.

The analysis confirms that the business is a high-frequency, low-basket-value grocery operation built around fresh, perishable staples. Produce and dairy items — bananas, organic strawberries, spinach, avocados, and milk — dominate both order volume and reorder rate, despite representing a comparatively small share of the total product catalogue. Ordering activity is heavily concentrated between 9 a.m. and 5 p.m., peaking mid-morning, and Sunday–Monday together account for a disproportionate share of weekly volume. Reorder behaviors are not uniform: perishable, routine-replenishment categories (milk, eggs, fresh fruit, water) show reorder rates 3–4 times higher than discretionary categories such as beauty, spices, or kitchen supplies.

Cross-referencing SQL validation, Python behavioral analysis, and the four Power BI dashboards produced a consistent, triangulated view of the customer base: a small set of high-frequency, high-loyalty SKUs carries a disproportionate share of demand, while the majority of the 49,355-product catalogue sees minimal repeat engagement. This 'few SKUs, most demand' pattern, combined with clear time-of-day and day-of-week seasonality, forms the basis of the recommendations in this report — spanning promotional timing, replenishment-based marketing (e.g. subscribe-and-save on top reorder items), staffing and delivery-slot allocation, and catalogue rationalization.

Report at a Glance

63,100 customers · 49,355 products · 134 aisles · 21 departments · ~10.5 items per order · 59.85% average reorder rate. Peak ordering window: 9 a.m.–5 p.m., highest volume Sunday–Monday.

Project Background

Online grocery platforms generate large volumes of transactional data every day — what was ordered, when, by whom, how it was categorized, and whether it was purchased again. On its own, this data is simply a log of activity. Its business value only emerges when it is validated, structured, and analyzed to reveal repeatable patterns in customer behavior and product performance.

This project works with a relational dataset (`project_orders`) modelled on the well-known Instacart Market Basket dataset structure, comprising five linked tables: `aisles`, `departments`, `products`, `orders`, and `order_products_train`. The `order_products_train` table maps products to the subset of orders flagged with `eval_set = "train"`, which is the labelled, analysis-ready slice of the full order history. Together these tables allow every product purchased to be traced back to its aisle, its department, the customer who bought it, and the date/time context of the order.

The engagement was structured in three stages, mirroring how a real analytics team would typically operate: (1) MySQL was used to connect to the hosted database, validate data integrity, and perform first-pass exploration; (2)

Python (via Google Colab) was used to merge the five tables into a single analytical dataset and answer 20 specific business questions with supporting statistical analysis; and (3) Power BI was used to convert the processed output into four interactive executive dashboards for non-technical stakeholders.

Business Problem

Grocery e-commerce operates on thin margins and high order frequency, which means small improvements in basket size, reorder rate, or delivery-slot efficiency compound into meaningful revenue impact. Without a structured view of purchasing behavior, Marketing and Merchandising teams are left to make decisions — which products to promote, which time slots to priorities, which customers are at risk of churn — based on intuition rather than evidence.

The core business problem addressed in this project is: **how can historical order data be used to optimize marketing strategy and improve customer satisfaction?** This requires answering three linked questions that recur throughout this report:

- What happened? — What are the dominant patterns in product demand, reorder behavior, and order timing?
- Why did it happen? — What structural or behavioral factors explain those patterns (perishability, routine replenishment, catalogue depth, weekly shopping rhythms)?
- What should stakeholders do? — Which concrete, data-backed actions should Marketing, Merchandising, and Operations take as a result?

Project Objective

The project brief specified 20 analytical questions spanning product catalogue structure, customer behavior, order timing, and reorder patterns. Rather than treating these as 20 isolated queries, they were grouped into four analytical objectives that map directly to business decisions:

- **Catalogue & Assortment Objective** — quantify how products are distributed across departments and aisles, and identify which categories carry the most SKUs versus the most demand (Qs 1–3, 20).
- **Customer Behavior Objective** — profile the customer base by order frequency, basket size, department reach, and purchase cadence (Qs 5, 6, 10, 11, 14, 19).
- **Temporal / Demand-Timing Objective** — identify peak ordering hours and days, and how basket size shifts across the week, to inform staffing and promotional scheduling (Qs 7, 8, 15, 16, 18).
- **Product Performance & Loyalty Objective** — surface the products and aisles with the highest reorder rates, and the categories that anchor repeat purchase behavior, to guide marketing and retention strategy (Qs 4, 9, 12, 13, 17).

Each objective is addressed using the same discipline: SQL for validation, Python for quantitative analysis, and Power BI for stakeholder-facing visualization — with every finding pushed one step further than "what happened" to explain why it matters and what action it implies.

Dataset Overview

The analysis is built on a five-table relational schema hosted in a MySQL database (project_orders). The tables and their validated row counts, confirmed independently in both SQL and Python, are summarized below.

Data Summary

Table	Description	Records
Aisles	Aisle reference list (aisle_id → aisle name)	134
Departments	Department reference list (department_id → department name)	21
Products	Product catalogue (product_id, name, aisle_id, department_id)	49355
Orders	One row per order (user_id, eval_set, order timing fields)	1,048,575
Order Products (Train)	One row per product within an order (order_id, product_id, cart position, reordered flag)	1,048,575

Two structural points from the dataset were carried through into the analysis rather than treated as incidental detail:

Train-set scope: the order_products table is explicitly named order_products_train and only maps to orders where eval_set = "train". This is a deliberate, labelled analytical subset (consistent with the standard Instacart dataset design), not the full order history. After merging, the analytical base table (master_df) contains 318,051 order-product line items. All findings in this report describe patterns within this train-set slice.

Row-count ceiling: both the orders and order_products_train tables report exactly 1,048,575 rows — one short of 2^{20} (1,048,576), the well-known row-import ceiling of tools such as Excel/CSV-based bulk loaders. This strongly suggests the source tables were truncated at import rather than reflecting a naturally occurring row count. It does not affect the validity of the patterns identified, but it is noted here as a data-limitation caveat for anyone extending this analysis.

The tables are connected using primary and foreign keys:

- **order_id** connects Orders and Order Products.
- **product_id** connects Order Products and Products.
- **aisle_id** connects Products and Aisles.
- **department_id** connects Products and Departments.

These relationships allow comprehensive analysis by combining customer, product, and order information into a unified analytical dataset.

Data Quality Result

Independent validation in both SQL and Python confirmed zero missing values and zero duplicate primary keys across all five tables, and referential integrity held across every join (products ↔ aisles, products ↔ departments, orders ↔ order_products). The dataset is clean and analysis-ready. One notable exception is discussed in the SQL Analysis section: an aisle explicitly labelled "missing" (aisle_id 100) containing 1,254 products — a real category value, not a null.

Tools Used and Methodology

Tools Used

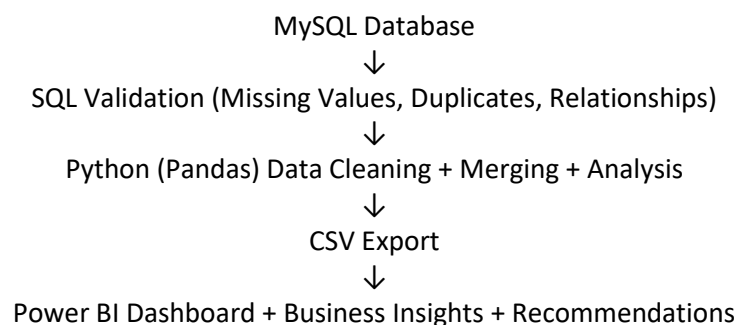
Layer	Table	Purpose in this Project
Database	MySQL (hosted, project_orders)	Source of truth; data validation and first-pass exploration
Analysis	Python (Google Colab): Pandas, NumPy	Table merging, aggregation, and the 20-question analysis
Visualization (analysis)	Matplotlib	Exploratory charts (peak hours, day-of-week volume)
Visualization (reporting)	Microsoft Power BI	Interactive executive dashboards
Data interchange	CSV (UTF-8)	Bridge format between the Python analysis layer and Power BI

Methodology

The project followed a standard end-to-end analytics pipeline, moving from raw relational data to validated analysis to stakeholder-facing reporting:

1. **Connect & Explore (SQL):** Connected directly to the hosted MySQL instance, inspected schema structure (SHOW TABLES, DESCRIBE), and captured row counts for all five tables.
2. **Validate (SQL):** Checked every table for missing values, duplicate primary keys, and referential integrity via join tests, before any analysis was trusted.
3. **Load & Merge (Python):** Connected to the same database from Python using SQLAlchemy/PyMySQL, re-verified data quality independently, filtered orders to eval_set = "train", and merged order_products with orders, products, aisles, and departments into a single analytical table (master_df, 318,051 rows × 15 columns).
4. **Analyze (Python):** Answered all 20 brief questions using groupby/aggregation logic, producing both summary statistics and two supporting exploratory charts (peak order hours, order volume by day of week).
5. **Export (Python → CSV):** Exported the cleaned source tables and key derived results (e.g. master_df, top-reorder-product tables, department-level reorder summaries) to CSV, which serves as the interchange layer into the visualization tool.
6. **Visualize (Power BI):** Imported the CSV outputs into Power BI and built four purpose-built dashboards — Executive Business Analysis, Customer Business Analysis, Product Performance Analysis, and Department & Category Analysis — for different stakeholder audiences.

Data Pipeline



Database Exploration and Validation using SQL

Before any behavioral analysis was performed, the raw dataset was validated directly in MySQL. This stage exists to answer a single question that every later insight depends on: can this data be trusted? The SQL queries were grouped into purposeful categories rather than run and reported as a flat list.

Database Connection

The analysis began by connecting to the MySQL database provided for the project. The database contained five interconnected tables representing products, orders, departments, aisles, and customer purchase history. SQL was used as the primary tool to understand the database structure and verify data integrity before performing analytical tasks. To confirm a successful connection and verify the available schemas, the following query or command was executed.

```
SHOW DATABASES;  
USE project_order;
```

Result Grid:

information_schema
project_orders

Database Exploration

A preliminary examination of the dataset was performed to determine the size of each table and to understand the overall volume of data available for analysis. Row counts were obtained for all tables, and sample records were inspected using SQL queries to verify that the imported data was complete and correctly structured.

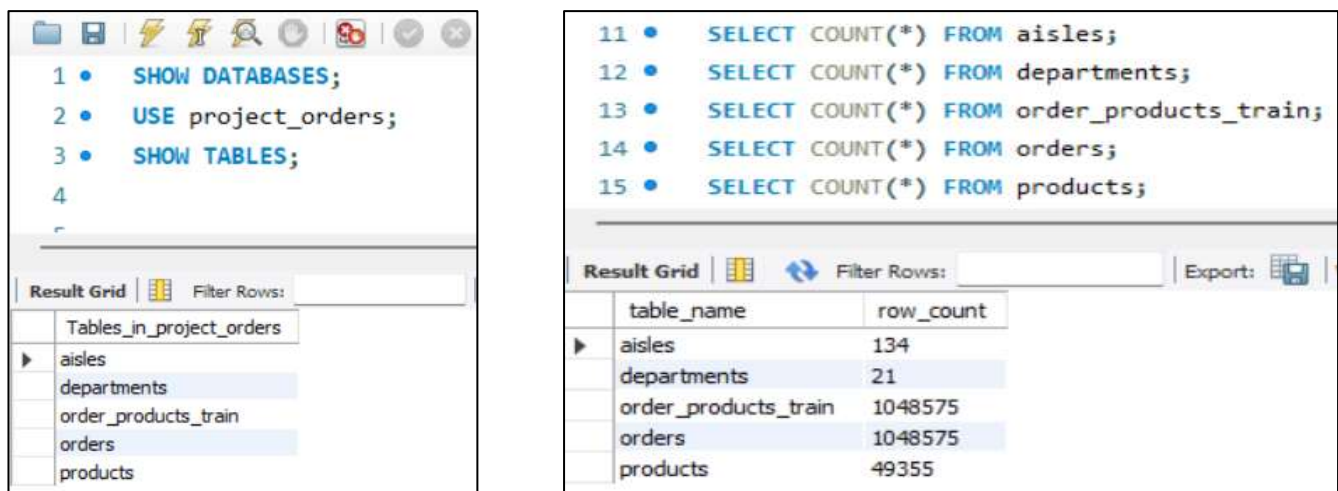


Figure 1: Database exploration queries and output grids.

The database schema was explored to understand the available tables, their attributes, and relationships. Commands such as `SHOW TABLES` and `DESCRIBE` were executed to identify the structure of each table, primary keys, and the data types of every attribute. This step provided a clear understanding of how different tables were connected and prepared the foundation for further analysis.

5 DESCRIBE aisles;
6 DESCRIBE departments;
7 DESCRIBE order_products_train;
8 DESCRIBE orders;
9 • DESCRIBE products;

Result Grid

Filter Rows:

Export:

Wrap Cell Contents:

	TABLE_NAME	COLUMN_NAME	DATA_TYPE	IS_NULLABLE	COLUMN_KEY	COLUMN_DEFAULT	EXTRA
▶	aisles	aisle_id	int	NO	PRI	NULL	
	aisles	aisle	varchar	NO		NULL	
	departments	department_id	int	NO	PRI	NULL	
	departments	department	varchar	NO		NULL	
	order_products_train	order_id	int	NO		NULL	
	order_products_train	product_id	int	NO		NULL	
	order_products_train	add_to_cart_order	int	NO		NULL	
	order_products_train	reordered	tinyint	NO		NULL	
	orders	order_id	int	NO	PRI	NULL	
	orders	user_id	int	YES		NULL	
	orders	eval_set	varchar	YES		NULL	
	orders	order_number	int	YES		NULL	
	orders	order_dow	int	YES		NULL	
	orders	order_hour_of_day	int	YES		NULL	
	orders	days_since_prior...	varchar	YES		NULL	
	products	product_id	int	NO	PRI	NULL	
	products	product_name	text	YES		NULL	
	products	aisle_id	int	YES		NULL	
	products	department_id	int	YES		NULL	

Figure 2: SQL queries utilized to inspect table schemas and the resulting column metadata grid

Database Summary Table:

Table	Row Count	Primary Key
aisles	134	aisle_id
departments	21	department_id
products	49,355	product_id
orders	1,048,575	order_id
order_products_train	1,048,575	order_id + product_id (composite)

Exploration Summary

Primary Keys were unique: Every table's designated primary key (such as product_id, order_id, and aisle_id) was verified to ensure it contained entirely unique values with no overlaps. This guarantees that each entity—whether it is a product, an order, or a department—can be distinctly and reliably identified without ambiguity.

Table Relationships were consistent: The foreign keys mapping across different tables (for example, linking a product_id in the transactional data back to the master products table) were completely aligned. This means there are no orphaned records or broken links, ensuring perfect referential integrity for deep-dive analysis.

Sample Data Verification

To understand the structure and contents of the database, sample records from each table were retrieved using the `SELECT * FROM <table> LIMIT 5` command. This preliminary exploration helped verify the available attributes, data format, and relationships between tables before performing analytical queries.

The following tables were explored:

- **Aisles:** Contains the unique aisle identifiers and corresponding aisle names.
- **Departments:** Contains the department identifiers and department names.
- **Products:** Stores product information along with the associated aisle and department identifiers.
- **Orders:** Contains customer order details, including user ID, order sequence, order day, order hour, and days since the previous order.
- **Order_Products_Train:** Represents the relationship between orders and products, including the sequence in which products were added to the cart and whether they were reordered.

17 • SELECT * FROM aisles LIMIT 5;

Result Grid

Filter Rows:

	aisle_id	aisle
▶	1	prepared soups salads
	2	specialty cheeses
	3	energy granola bars
	4	instant foods
	5	marinades meat preparation
*	NULL	NULL

19 • SELECT * FROM products LIMIT 5;

Result Grid

Filter Rows:

Edit:

Export/Import:

	product_id	product_name	aisle_id	department_id
▶	1	Chocolate Sandwich Cookies	61	19
	2	All-Seasons Salt	104	13
	3	Robust Golden Unsweetened Oolong Tea	94	7
	4	Smart Ones Classic Favorites Mini Rigatoni With ...	38	1
	5	Green Chile Anytime Sauce	5	13
*	NULL	NULL	NULL	NULL

21 • SELECT * FROM order_products_train LIMIT 5;

Result Grid

Filter Rows:

Export:

Wrap

	order_id	product_id	add_to_cart_order	reordered
▶	1	49302	1	1
	1	11109	2	1
	1	10246	3	0
	1	49683	4	0
	1	43633	5	1

18 • SELECT * FROM departments LIMIT 5;

Result Grid

Filter Rows:

Edit:

	department_id	department
▶	1	frozen
	2	other
	3	bakery
	4	produce
	5	alcohol
*	NULL	NULL

20 • SELECT * FROM orders LIMIT 5;

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell

	order_id	user_id	eval_set	order_number	order_dow	order_hour_of_day	days_since_prior_order
▶	6	22352	prior	4	1	12	30
	8	3107	prior	5	4	6	17
	13	45082	prior	2	6	17	1
	14	18194	prior	49	3	15	3
	15	54901	prior	51	3	11	2
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 3: Sample Records Retrieved from the Five Database Tables

The sample records confirmed that the database structure matched the provided schema and that all tables contained valid and consistent data. This verification ensured that the dataset was ready for table joins and further analytical processing.

Data Validation

To ensure the utmost reliability and accuracy of the subsequent analysis, a series of rigorous data quality checks were systematically performed. The entire dataset underwent a thorough examination to guarantee it met the necessary standards for robust business intelligence, effectively eliminating the need for any additional pre-processing or data cleaning.

Missing Value Analysis: Every column across all five tables was scanned using targeted aggregation queries to detect null or missing values. **Result:** No Missing Values

- Checked all columns across all five tables for NULL values.
- No missing values were found in the dataset.

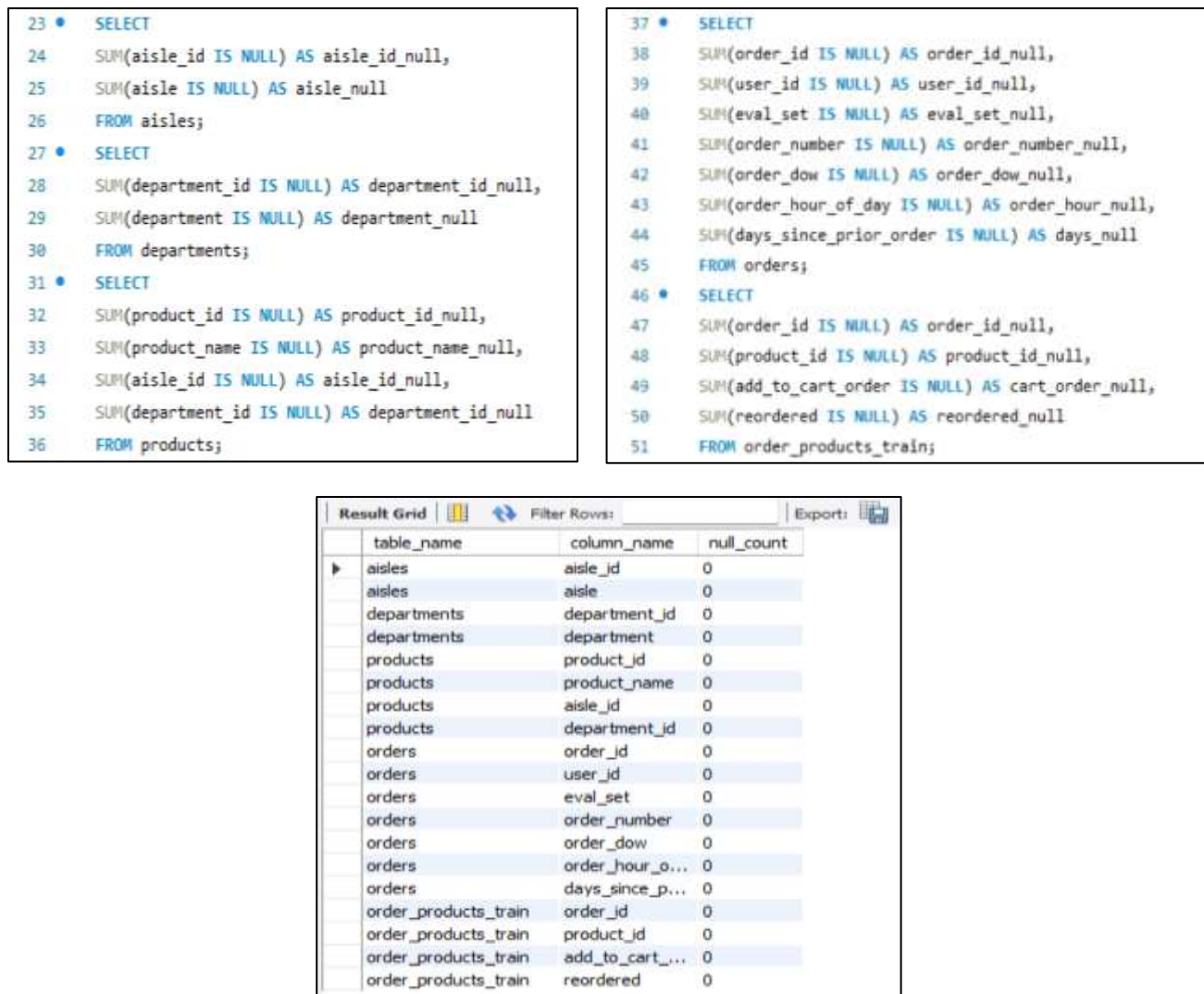
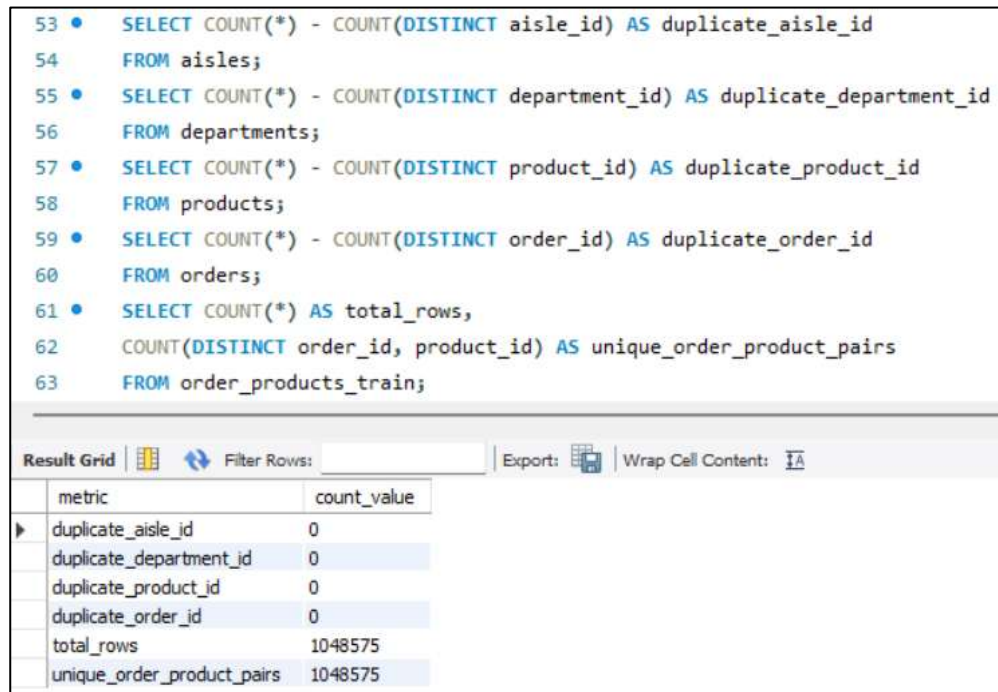


Figure 4: SQL queries used to aggregate and verify the count of null or missing values across all database tables, confirming a completely clean dataset.

Duplicate Record Verification: To prevent skewed aggregations and duplicate counting during analysis, an audit was conducted to identify any redundant rows or broken primary key constraints. **Result:** No Duplicate Records

- Verified duplicate records using Primary Keys and appropriate key combinations.
- No duplicate records were identified.



The screenshot shows a SQL IDE with a query editor and a results grid. The query editor contains the following SQL code:

```
53 • SELECT COUNT(*) - COUNT(DISTINCT aisle_id) AS duplicate_aisle_id
54 FROM aisles;
55 • SELECT COUNT(*) - COUNT(DISTINCT department_id) AS duplicate_department_id
56 FROM departments;
57 • SELECT COUNT(*) - COUNT(DISTINCT product_id) AS duplicate_product_id
58 FROM products;
59 • SELECT COUNT(*) - COUNT(DISTINCT order_id) AS duplicate_order_id
60 FROM orders;
61 • SELECT COUNT(*) AS total_rows,
62 COUNT(DISTINCT order_id, product_id) AS unique_order_product_pairs
63 FROM order_products_train;
```

The results grid shows the following data:

metric	count_value
duplicate_aisle_id	0
duplicate_department_id	0
duplicate_product_id	0
duplicate_order_id	0
total_rows	1048575
unique_order_product_pairs	1048575

Figure 5: SQL queries used to check for duplicate records across the tables, along with the execution results

Database Validation Summary

All validation checks confirmed that the dataset is clean, consistent, and suitable for further analysis. Since no missing values or duplicate records were found, no additional data cleaning was required before proceeding with SQL and Python analysis.

Table Relationship Validation

Before beginning the business analysis, SQL JOIN operations were performed to validate the relationships among the five tables. Products were successfully linked with aisles and departments, while customer orders were connected with purchased products. These joins confirmed that the database relationships were accurate and enabled the creation of a unified analytical dataset.

The following relationships were verified:

- `Products` ↔ `Aisles` using `aisle_id`
- `Products` ↔ `Departments` using `department_id`
- `Orders` ↔ `Order_Products_Train` using `order_id`
- **Complete Database Join** combining Orders, Products, Aisles, Departments, and Order Products into a single analytical view

```

65 • SELECT p.product_name, a.aisle
66 FROM products p JOIN aisles a
67 ON p.aisle_id = a.aisle_id LIMIT 10;

```

product_name	aisle
Chocolate Sandwich Cookies	cookies cakes
All-Seasons Salt	spices seasonings
Robust Golden Unsweetened Oolong Tea	tea
Smart Ones Classic Favorites Mini Rigatoni With ...	frozen meals
Green Chile Anytime Sauce	marinades meat preparation
Dry Nose Oil	cold flu allergy
Pure Coconut Water With Orange	juice nectars
Cut Russet Potatoes Steam N' Mash	frozen produce
Light Strawberry Blueberry Yogurt	yogurt
Sparkling Orange Juice & Priddy Pear Beverage	water seltzer sparkling water

```

69 • SELECT p.product_name, d.department
70 FROM products p JOIN departments d
71 ON p.department_id = d.department_id LIMIT 10;

```

product_name	department
Chocolate Sandwich Cookies	snacks
All-Seasons Salt	pantry
Robust Golden Unsweetened Oolong Tea	beverages
Smart Ones Classic Favorites Mini Rigatoni With ...	frozen
Green Chile Anytime Sauce	pantry
Dry Nose Oil	personal care
Pure Coconut Water With Orange	beverages
Cut Russet Potatoes Steam N' Mash	frozen
Light Strawberry Blueberry Yogurt	dairy eggs
Sparkling Orange Juice & Priddy Pear Beverage	beverages

```

77 • SELECT o.order_id, o.user_id,
78 p.product_name, a.aisle,
79 d.department, op.reordered
80 FROM order_products_train op
81 JOIN products p ON op.product_id = p.product_id
82 JOIN aisles a ON p.aisle_id = a.aisle_id
83 JOIN departments d ON p.department_id = d.department_id
84 JOIN orders o ON op.order_id = o.order_id LIMIT 10;

```

order_id	user_id	product_name	aisle	department	reordered
38	42756	Shelled Pistachios	nuts seeds dried fruit	snacks	0
38	42756	Organic Biologique Limes	packaged vegetables fruits	produce	0
38	42756	Organic Raw Unfiltered Apple Cider Vinegar	oils vinegars	pantry	0
38	42756	Organic Baby Arugula	packaged vegetables fruits	produce	1
38	42756	Organic Hot House Tomato	fresh vegetables	produce	0
38	42756	Green Peas	frozen produce	frozen	0
38	42756	Bunched Cilantro	fresh herbs	produce	0
38	42756	Flat Parsley, Bunch	fresh herbs	produce	0
38	42756	Fresh Dill	fresh herbs	produce	0
96	17227	Roasted Turkey	lunch meat	deli	1

```

73 • SELECT o.order_id, op.product_id
74 FROM orders o JOIN order_products_train op
75 ON o.order_id = op.order_id LIMIT 10;

```

order_id	product_id
38	11913
38	18159
38	4461
38	21616
38	23622
38	32433
38	28842
38	42625
38	39693
96	20574

Figure 7: Validation of Table Relationships Using SQL JOIN Operations

The successful execution of these JOIN operations confirmed that the database schema was correctly linked, ensuring that customer orders, products, aisles, and departments could be analyzed together without inconsistencies.

Python-Based Data Analysis

Data Preparation in Python

After completing the database exploration and validation using SQL, the analysis was shifted to Python to efficiently handle large datasets and perform advanced data manipulation. Python was chosen because it provides greater flexibility for data processing, aggregation, and analysis compared to SQL, especially when working with over one million records.

Before starting the analytical tasks, the following preprocessing steps were performed:

- Installed the required Python libraries.
- Imported all necessary libraries, including Pandas, NumPy, and SQL Alchemy.
- Established a connection to the MySQL database.
- Imported all five tables into Pandas Data Frames.
- Verified the successful loading of each table by checking their dimensions.
- Verified column names, data types, missing values, and duplicate records.
- Merged the five tables into a single master dataset using the appropriate primary and foreign keys.
- Confirmed that the merged dataset was suitable for further analysis.

After completing the preprocessing stage, the business questions were solved using Python and Pandas.

Query 1 – Top 10 Aisles with the Highest Number of Products

Objective: To identify the aisles containing the highest number of products and understand how products are distributed across different store aisles.

```
q1_top_aisles = (  
    products  
    .merge(aisles, on="aisle_id")  
    .groupby("aisle", as_index=False)  
    .agg(total_products=("product_id", "count"))  
    .sort_values("total_products", ascending=False))  
print(show(q1_top_aisles, 10))
```

Additional verification: verifying what "missing" actually means

```
aisles[aisles["aisle"] == "missing"]
```

	aisle	total_products		aisle_id	aisle
0	missing	1254			
1	candy chocolate	1235			
2	ice cream ice	1084			
3	vitamins supplements	1030			
4	yogurt	1023			
5	chips pretzels	980			
6	packaged cheese	888			
7	tea	887			
8	frozen meals	880			
9	cookies cakes	869	99	100	missing

Interpretation

- The **"missing"** aisle contains the highest number of products (1,254).
- Candy Chocolate, Ice Cream, Vitamins & Supplements, and Yogurt also contain a large variety of products.
- The distribution indicates that certain aisles have a significantly broader product assortment than others.
- Verification confirmed that "missing" is an actual aisle name (aisle_id = 100) rather than missing or incomplete data.

Business Insight

Aisles with a larger product assortment require improved inventory planning, shelf management, and category organization. High-product aisles should receive greater attention during inventory replenishment and promotional planning.

Query 2 – Total Number of Unique Departments

Objective: To determine the total number of unique departments available in the dataset. This provides an overview of the major product categories offered by the store and helps understand the overall product classification.

```
q2_department_count = departments["department"].nunique()
print(f"Total Unique Departments: {q2_department_count}")
```

Total Unique Departments: 21

Interpretation

- The dataset contains **21 unique departments**, representing the major product categories available.
- Each department groups together products with similar characteristics, making it easier to organize inventory and analyse product performance.
- This confirms that the store offers a wide variety of product categories to meet different customer needs.

Business Insight

The presence of **21 departments** indicates a diverse product portfolio. Department-level analysis can help management identify high-performing and low-performing categories, optimize inventory allocation, and design targeted marketing strategies for different customer segments.

Query 3 – Distribution of Products Across Departments

Objective: To analyse how products are distributed across different departments and identify the departments with the largest product assortment.

```
q3_department_distribution = (
    products
    .merge(departments, on="department_id")
    .groupby("department", as_index=False)
    .agg(total_products=("product_id", "count"))
    .sort_values("total_products", ascending=False))
show(q3_department_distribution)
```

	department	total_products
0	personal care	6524
1	snacks	6221
2	pantry	5331
3	beverages	4324
4	frozen	3982
5	dairy eggs	3428
6	household	3068
7	canned goods	2073
8	dry goods pasta	1845
9	produce	1671

Interpretation

- Personal Care has the largest product assortment with 6,524 products, followed closely by Snacks and Pantry.
- Departments such as Produce, Dry Goods Pasta, and Canned Goods contain comparatively fewer products.
- The distribution of products across departments is uneven, indicating that certain departments offer a much broader variety of products than others.

Business Insight

The product catalogue is concentrated in a few major departments, particularly Personal Care, Snacks, and Pantry. These departments require greater attention in inventory planning, category management, and shelf space allocation due to their extensive product range.

Query 4 – Top 10 Products with the Highest Reorder Rates

Objective: To identify the products with the highest customer reorder rates and evaluate customer loyalty towards individual products. Since products with very few orders can produce misleadingly high reorder rates, different minimum order thresholds were applied to ensure the reliability of the analysis.

```
def top_reorder_products(min_orders):
    result = (
        master_df
        .groupby("product_name", as_index=False)
        .agg(
            total_orders=("product_id", "count"),
            reorder_count=("reordered", "sum")
        )
    )
    result["reorder_rate"] = (
        result["reorder_count"] / result["total_orders"] * 100
    ).round(2)
    result = (
        result[result["total_orders"] >= min_orders]
        .sort_values(
            ["reorder_rate", "total_orders"],
            ascending=[False, False]
        )
        .head(10)
    )
    return result
print("Minimum Orders = 20")
display(top_reorder_products(20))
print("Minimum Orders = 50")
display(top_reorder_products(50))
print("Minimum Orders = 100")
display(top_reorder_products(100))
```

Minimum Orders = 20

	product_name	total_orders	reorder_count	reorder_rate
658	A+ Cinnamon Sweet Potato Pancakes	21	21	100.00
12449	Maple Honey Ham	23	22	95.65
22070	Sparkling Water, Bottles	20	19	95.00
22895	Super Natural Organic Whole Milk	20	19	95.00
15943	Organic Low Fat Milk	100	93	93.00
15859	Organic Lactose Free 1% Lowfat Milk	68	63	92.65
5762	Crinkle Cut French Fries	27	25	92.59
25670	Wild Berry Smoothie	50	46	92.00
7975	French Vanilla Coffee Creamer	25	23	92.00
25908	Yogurt Tubes Blueberry	21	19	90.48

Minimum Orders = 50

	product_name	total_orders	reorder_count	reorder_rate
15943	Organic Low Fat Milk	100	93	93.00
15859	Organic Lactose Free 1% Lowfat Milk	68	63	92.65
25670	Wild Berry Smoothie	50	46	92.00
17136	Organic Whole Milk with DHA Omega-3	115	104	90.43
13030	Milk, Organic, Vitamin D	139	124	89.21
16464	Organic Reduced Fat Milk	274	242	88.32
15961	Organic Lowfat 1% Milk	93	82	88.17
1809	Banana	4357	3840	88.13
8013	Fresh CA Grown Eggs	92	81	88.04
32	1% Lowfat Milk	99	87	87.88

Minimum Orders = 100

	product_name	total_orders	reorder_count	reorder_rate
15943	Organic Low Fat Milk	100	93	93.00
17136	Organic Whole Milk with DHA Omega-3	115	104	90.43
13030	Milk, Organic, Vitamin D	139	124	89.21
16464	Organic Reduced Fat Milk	274	242	88.32
1809	Banana	4357	3840	88.13
19424	Pure Sparkling Water	258	222	86.05
1689	Bag of Organic Bananas	3522	3016	85.63
17126	Organic Whole Milk	1121	957	85.37
15396	Organic Fat Free Milk	188	160	85.11
15864	Organic Large Brown Grade AA Cage Free Eggs	257	216	84.05

Why the Threshold of 100 Was Selected

A threshold of **100 minimum orders** was chosen because it provides a more reliable representation of customer purchasing behaviour. Lower thresholds (20 and 50) included products with limited purchase histories, which could produce inflated reorder rates due to small sample sizes. Using a higher threshold ensures that the reported reorder rates are based on a substantial number of customer purchases, making the analysis more suitable for business decision-making.

Interpretation

- Products with very low order volumes can achieve artificially high reorder rates, making them less reliable indicators of customer loyalty.
- Comparing multiple thresholds demonstrated that increasing the minimum order requirement filters out statistically weak results.
- Organic milk products consistently ranked among the highest reordered products across all thresholds, indicating strong and stable customer loyalty.
- Banana and Bag of Org. Bananas remained among the top reordered products despite having thousands of total orders, confirming that their popularity is supported by a large cust base rather than a small sample.

Business Insight

To ensure meaningful business decisions, a **minimum order threshold of 100** was selected for the final analysis. This approach balances sample size with statistical reliability and prevents products with very few purchases from dominating the rankings. The findings indicate that **organic dairy products, bananas, and sparkling water** are highly loyal products that should be prioritized for inventory planning, promotional campaigns, and customer retention strategies.

Query 5 – Total Number of Unique Users

Objective: To determine the total number of unique customers who have placed orders in the dataset. This helps understand the size of the active customer base and provides a foundation for customer behaviour analysis.

```
q5_unique_users = orders["user_id"].nunique()
print(f"Total Unique Users: {q5_unique_users}")
```

Total Unique Users: 63100

Interpretation

- The dataset contains **63,100 unique customers** who have placed orders.
- This indicates a large and diverse customer base, making the dataset suitable for customer behaviour and purchasing pattern analysis.
- A larger customer base improves the reliability of analytical insights and business decision-making.

Business Insight

The presence of over **63,000 unique customers** indicates a broad market reach. This provides opportunities for customer segmentation, personalized marketing campaigns, loyalty programs, and targeted promotional strategies to improve customer retention and increase sales.

Query 6 – What is the Average Number of Days Between Orders for Each User?

Objective: To calculate the average number of days between consecutive orders for each customer. This helps understand customer purchasing frequency and provides insights into shopping behaviour and customer retention.

```
q6_avg_days_between_orders = (
    orders[orders["days_since_prior_order"].notna()]
    .copy()
    .assign(days_since_prior_order=lambda df: pd.to_numeric(df["days_since_prior_order"]))
    .groupby("user_id", as_index=False)
    .agg(avg_days_between_orders=("days_since_prior_order", "mean"))
    show(q6_avg_days_between_orders)
```

	user_id	avg_days_between_orders
0	1	19.000000
1	2	16.285714
2	3	12.000000
3	4	17.000000
4	5	11.500000
5	6	13.333333
6	7	10.450000
7	8	23.333333
8	9	22.000000
9	10	21.800000

Interpretation

- The average number of days between orders varies across customers, indicating different purchasing habits.
- Some customers place orders approximately every **10–12 days**, while others wait **20 days or more** before making another purchase.
- This variation suggests the presence of both frequent and occasional shoppers within the customer base.

Business Insight

Customer purchase intervals provide valuable information about buying behavior. Customers with shorter ordering intervals are generally more engaged and may respond well to loyalty programs, while customers with longer intervals can be targeted through personalized reminders and promotional offers to encourage repeat purchases.

Query 7 – Peak Hours of Order Placement During the Day

Objective: To identify the hours of the day during which customers place the highest number of orders. This analysis helps understand customer purchasing patterns and supports better operational planning.

```
q7_peak_order_hours = (  
    orders  
    .groupby("order_hour_of_day", as_index=False)  
    .agg(total_orders=("order_id", "count"))  
    .sort_values("total_orders", ascending=False))  
print(q7_peak_order_hours)  
# Visualization  
import matplotlib.pyplot as plt  
plt.figure(figsize=(10,5))  
plt.bar(  
    q7_peak_order_hours["order_hour_of_day"],  
    q7_peak_order_hours["total_orders"])  
plt.title("Orders by Hour of Day")  
plt.xlabel("Hour of Day")  
plt.ylabel("Number of Orders")  
plt.show()
```

	order_hour_of_day	total_orders
10	10	88228
11	11	87087
14	14	86905
15	15	86888
13	13	85652
12	12	84204
16	16	83421
9	9	78737
17	17	69960
18	18	56284
8	8	54868
19	19	43076
20	20	31375
7	7	28199
21	21	23972
22	22	18824
23	23	12112
6	6	9253
0	0	6881
1	1	3698
5	5	2882
2	2	2312
4	4	1748
3	3	1649

Interpretation

- Customer ordering activity is highest between **10:00 AM and 4:00 PM**.
- The single busiest hour is **10:00 AM**, with **88,228 orders**.
- Order volume gradually decreases after **4:00 PM**.
- The lowest ordering activity occurs during the early morning hours (**12:00 AM to 5:00 AM**), when customer activity is minimal.

Business Insight

Customer demand is concentrated during the late morning and afternoon, indicating predictable shopping behaviour. This information can be used to optimize staffing levels, warehouse operations, inventory replenishment schedules, and delivery planning during peak demand periods.



Query 8 – How Does Order Volume Vary by Day of the Week?

Objective: To analyse the distribution of customer orders across different days of the week and identify the busiest shopping days. This helps understand customer purchasing patterns and supports better inventory planning and workforce management.

```
import matplotlib.pyplot as plt
q8_orders_by_day = (
    orders
    .groupby("order_dow", as_index=False)
    .agg(total_orders=("order_id", "count"))
    .sort_values("order_dow")
)
display(q8_orders_by_day)
plt.figure(figsize=(8,5))
plt.bar(
    q8_orders_by_day["order_dow"],
    q8_orders_by_day["total_orders"])
plt.title("Order Volume by Day of Week")
plt.xlabel("Day of Week (0 = Sunday)")
plt.ylabel("Total Orders")
plt.xticks(range(7))
plt.tight_layout()
plt.show()
```

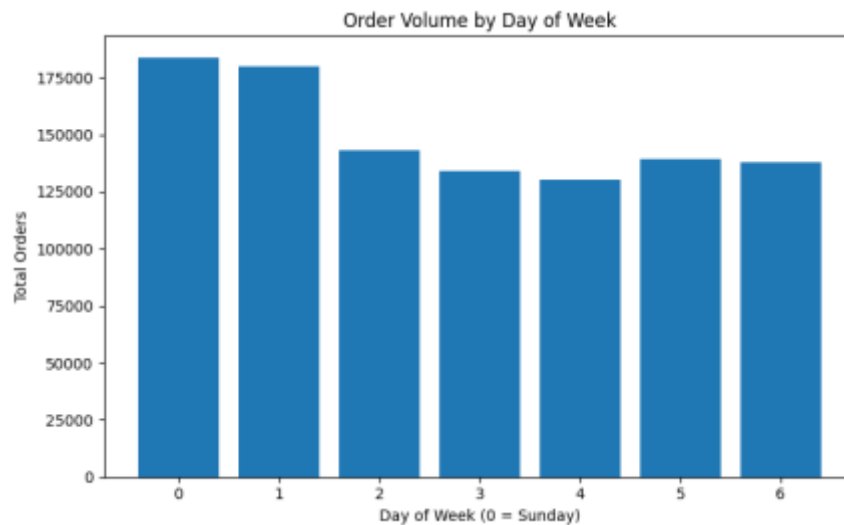
	order_dow	total_orders
0	0	183939
1	1	180025
2	2	143162
3	3	133839
4	4	130367
5	5	139183
6	6	138060

Interpretation

- The highest number of orders was placed on **Sunday (183,939)**, followed closely by **Monday (180,025)**.
- Order volume gradually decreases from Tuesday to Thursday.
- A slight increase in order volume is observed on Friday and Saturday.
- The results indicate that customer purchasing activity is concentrated at the beginning of the week.

Business Insight

The analysis suggests that **Sunday and Monday are the busiest shopping days**, making them critical periods for inventory availability and operational planning. The business should ensure adequate stock levels, workforce allocation, and delivery capacity during these peak days to meet customer demand efficiently. Promotional campaigns and product launches can also be strategically scheduled around these high-traffic days to maximize customer engagement and sales performance.



Query 9 – Top 10 Most Ordered Products

Objective: To identify the products that are purchased most frequently by customers. This analysis helps determine the highest-demand products and supports inventory planning, demand forecasting, and promotional strategies.

```
q9_top_ordered_products = (
    master_df
    .groupby("product_name", as_index=False)
    .agg(total_orders=("product_id", "count"))
    .sort_values("total_orders", ascending=False)
    .head(10)
)
display(q9_top_ordered_products)
```

	product_name	total_orders
1809	Banana	4357
1689	Bag of Organic Bananas	3522
16768	Organic Strawberries	2520
14733	Organic Baby Spinach	2286
11392	Large Lemon	1841
15689	Organic Hass Avocado	1705
14706	Organic Avocado	1661
22556	Strawberries	1479
11930	Limes	1383
16388	Organic Raspberries	1279

Interpretation

- **Banana** is the most frequently ordered product, with **4,357 orders**, making it the highest-demand item in the dataset.
- Fresh fruits and vegetables dominate the list, indicating that customers frequently purchase fresh produce.
- Organic products such as **Organic Bananas, Organic Strawberries, Organic Baby Spinach, and Organic Avocados** appear multiple times among the top-selling products, suggesting strong customer preference for organic food items.
- The top 10 products are primarily everyday grocery essentials that customers purchase regularly.

Business Insight

The analysis indicates that fresh produce and organic products are the primary demand drivers within the store. Maintaining adequate inventory levels for these high-demand products is essential to prevent stock shortages and improve customer satisfaction. Additionally, these products can be leveraged in promotional campaigns, bundle offers, and cross-selling strategies to increase customer engagement and overall sales.

Query 10 – Number of Users Who Have Placed Orders in Each Department

Objective: To identify the number of unique customers who have purchased products from each department. This analysis helps understand customer reach across different product categories and identifies the departments that attract the largest customer base.

```
q10_users_per_department = (
    master_df
    .groupby("department", as_index=False)
    .agg(unique_users=("user_id", "nunique"))
    .sort_values("unique_users", ascending=False))
display(q10_users_per_department)
```

	department	unique_users
19	produce	22303
7	dairy eggs	20153
3	beverages	14024
20	snacks	13304
10	frozen	11742
16	pantry	10886
2	bakery	8326
8	deli	7430
6	canned goods	6697
9	dry goods pasta	5942

13	meat seafood	5193
4	breakfast	5027
11	household	4897
17	personal care	3416
12	international	2188
14	missing	1630
1	babies	1435
0	alcohol	679
18	pets	622
15	other	392
5	bulk	323

Interpretation

- **Produce** has the highest customer reach, with **22,303 unique customers**, making it the most widely purchased department.
- **Dairy Eggs** follows closely with **20,153 unique customers**, indicating strong demand for everyday grocery essentials.
- **Beverages, Snacks, Frozen, and Pantry** also attract a large number of customers, highlighting their importance in customers' regular shopping baskets and Departments such as **Bulk, Other, Pets, and Alcohol** have comparatively fewer unique customers, indicating lower customer engagement.

Business Insight

The analysis shows that customers primarily purchase products from essential grocery departments such as **Produce, Dairy Eggs, Beverages, Snacks, and Frozen**. These departments should receive higher priority for inventory planning, promotional campaigns, and shelf space allocation, as they contribute to the largest customer reach. On the other hand, departments with lower customer engagement may require assortment optimization or targeted marketing strategies to improve their performance.

Query 11 – Average Number of Products per Order

Objective: To calculate the average number of products purchased in a single customer order. This helps in understanding the typical basket size and overall purchasing behaviour of customers.

```
q11_avg_products_per_order = (
    master_df
    .groupby("order_id")
    .size()
    .mean())
print(f"Average Products per Order: {q11_avg_products_per_order:.2f}")
```

Average Products per Order: 10.51

Interpretation

- The analysis shows that customers purchase an average of **10.51 products per order**.
- This indicates that most customers tend to buy multiple products in a single shopping session rather than making single-item purchases.
- The result suggests a moderate basket size, reflecting planned grocery shopping behaviour.

Business Insight

An average basket size of approximately **11 products** presents opportunities for cross-selling and upselling strategies. Retailers can increase the average order value by recommending complementary products, offering bundle discounts, or promoting frequently bought together items during checkout.

Query 12 – Most Reordered Product in Each Department

Objective: To identify the product with the highest number of repeat purchases in each department. This analysis helps determine the most popular and customer-loyal product within every product category.

```
q12_most_reordered = (  
    master_df  
    .groupby(["department", "product_name"], as_index=False)  
    .agg(reorder_count=("reordered", "sum"))  
q12_most_reordered = (  
    q12_most_reordered  
    .sort_values(  
        ["department", "reorder_count"],  
        ascending=[True, False]  
    )  
    .drop_duplicates("department")  
)  
display(q12_most_reordered)
```

	department	product_name	reorder_count
294	alcohol	Sauvignon Blanc	53
435	babies	Baby Food Stage 2 Blueberry Pear & Purple Carrot	44
943	bakery	100% Whole Wheat Bread	412
3921	beverages	Sparkling Water Grapefruit	637
4580	breakfast	Honey Nut Cheerios	167
4999	bulk	Dried Mango	78
5611	canned goods	Organic Black Beans	215
7737	dairy eggs	Organic Whole Milk	957
9086	deli	Original Hummus	539
9849	dry goods pasta	Marinara Sauce	117
10644	frozen	Blueberries	354
13040	household	100% Recycled Paper Towels	177
14812	international	Organic Sea Salt Roasted Seaweed Snacks	61
15300	meat seafood	Ground Turkey Breast	170
15894	missing	Organic Riced Cauliflower	89
16218	other	Roasted Almond Butter	19
16916	pantry	Extra Virgin Olive Oil	257
19838	personal care	Lavender Hand Soap	28
20962	pets	24/7 Performance Cat Litter	13
21467	produce	Banana	3840
24204	snacks	Lightly Salted Baked Snap Pea Crisps	164

Interpretation

- **Banana** is the most reordered product overall, with **3,840 repeat purchases**, making it the strongest-performing product in the Produce department.
- **Organic Whole Milk (957)** and **Sparkling Water Grapefruit (637)** also demonstrate exceptionally high customer loyalty within their respective departments.
- Departments such as **Pets**, **Alcohol**, and **Babies** have comparatively lower reorder counts, indicating lower repeat purchasing activity. Each department has a distinct product that customers reorder most frequently.

Business Insight

The results highlight the **flagship product** of each department. These products should be considered priority items for inventory management, promotional campaigns, and shelf placement, as they consistently drive repeat purchases and customer retention.

Query 13 – Products Reordered More Than Once

Objective: To identify how many products have been reordered more than once. This helps measure the number of products that encourage repeat purchases and indicates overall customer loyalty towards the product catalogue.

```
q13_products_reordered = (  
    master_df  
    .groupby("product_name")  
    .agg(reorder_count=("reordered", "sum"))  
q13_products_reordered = (  
    q13_products_reordered["reorder_count"] > 1  
).sum()  
print(f"Products Reordered More Than Once: {q13_products_reordered}")
```

Products Reordered More Than Once: 11529

Interpretation

- A total of **11,529 products** were reordered more than once by customers.
- This indicates that a significant portion of the product catalogue experiences repeat purchases rather than one-time sales.
- The result suggests that customers exhibit consistent purchasing behaviour across many products, demonstrating product loyalty within the marketplace.

Business Insight

A large number of repeatedly purchased products indicates a strong base of customer loyalty. The business should prioritize maintaining inventory for these products, monitor their availability closely, and use them in personalized recommendations, loyalty programs, and targeted promotional campaigns to maximize repeat sales and customer retention.

Query 14 – Average Number of Products Added to Cart per Order

Objective: To determine the average number of products added to the shopping cart in a single order. This helps understand the typical customer basket size and purchasing behaviour.

```
q14_avg_cart_size = (  
    master_df  
    .groupby("order_id")  
    .agg(cart_size=("add_to_cart_order", "max"))  
print(f"Average Cart Size: {q14_avg_cart_size['cart_size'].mean():.2f}")
```

Average Cart Size: 10.51

Interpretation

- On average, customers add approximately **10.51 products** to their shopping cart per order.
- The result reflects a moderate basket size, suggesting planned grocery shopping behaviour.
- **Since add_to_cart_order reaches the maximum item position in an order, both analyses produce the same business result (Query 11 and 14).**

Business Insight

The average cart size indicates opportunities for increasing basket value through cross-selling and product recommendations. Since customers already purchase multiple items in each order, introducing bundle offers, complementary product suggestions, or volume-based discounts could encourage customers to add even more products to their carts, thereby increasing the average order value and overall revenue.

Query 15 – How Does the Number of Orders Vary by Hour of the Day?

Objective: To identify how the number of customer orders is distributed across different hours of the day and determine the busiest shopping hours.

This query requires the same analysis performed in **Query 7 (Peak Hours of Order Placement During the Day)**. Since both questions analyse the distribution of orders across the 24-hour period, the same Python implementation and results have been reused to avoid duplication.

Reference: *Query 7 – Peak Hours of Order Placement During the Day.*

Query 16 – Distribution of Order Sizes

Objective: To analyse the distribution of customer order sizes by determining how many products customers purchase in each order. This helps understand typical shopping behaviour and identify the frequency of small, medium, and large orders.

```
q16_order_size_distribution = (  
    master_df  
    .groupby("order_id")  
    .agg(order_size=("product_id", "count"))  
    .groupby("order_size", as_index=False)  
    .agg(number_of_orders=("order_size", "count"))  
    .sort_values("order_size")  
)  
display(q16_order_size_distribution)
```

	order_size	number_of_orders
0	1	1536
1	2	1742
2	3	1883
3	4	1890
4	5	2032
...	...	...
62	66	1
63	67	1
64	70	1
65	72	1
66	74	1

Total Distinct Order Sizes: 67

Interpretation

- Customers place orders ranging from **1 to 74 products**.
- Smaller and medium-sized orders occur much more frequently than very large orders.
- As the order size increases, the number of orders gradually decreases.
- Orders containing more than **60 products** are extremely rare, with only one order recorded for several of the largest basket sizes.

Business Insight

The analysis indicates that most customers prefer purchasing a moderate number of products in a single order, while exceptionally large orders are uncommon. This purchasing pattern can help the business optimize inventory planning, warehouse operations, and order fulfillment processes. It also suggests an opportunity to increase the average basket size through personalized recommendations, product bundles, and cross-selling strategies.

Query 17 – Average Reorder Rate by Aisle

Objective: To identify the aisles with the highest average reorder rates and understand which product categories encourage the strongest repeat purchasing behaviour among customers.

```
q17_avg_reorder_rate = (  
    master_df  
    .groupby("aisle", as_index=False)  
    .agg(avg_reorder_rate=("reordered", "mean"))  
q17_avg_reorder_rate["avg_reorder_rate"] = (  
    q17_avg_reorder_rate["avg_reorder_rate"] * 100  
)  
.round(2)  
q17_avg_reorder_rate = (  
    q17_avg_reorder_rate  
    .sort_values("avg_reorder_rate", ascending=False))  
display(q17_avg_reorder_rate)
```

	aisle	avg_reorder_rate
83	milk	79.30
131	water seltzer sparkling water	73.97
50	fresh fruits	73.50
41	eggs	73.46
96	packaged produce	70.98
...	...	...
29	cold flu allergy	21.11
8	beauty	20.00
7	baking supplies decor	18.25
122	spices seasonings	17.65
76	kitchen supplies	16.33

Total Aisles Analyzed: 134

Interpretation

- The **Milk** aisle recorded the highest average reorder rate (**79.30%**), indicating that customers frequently repurchase products from this category.
- Other high-performing aisles, including **Water Seltzer & Sparkling Water**, **Fresh Fruits**, **Eggs**, and **Packaged Produce**, also demonstrated strong customer loyalty.
- In contrast, aisles such as **Kitchen Supplies**, **Spices & Seasonings**, **Baking Supp. Decor**, and **Beauty** exhibited comparatively lower reorder rates, suggesting that prod. in these categories are purchased less frequently.

Business Insight

The analysis indicates that **essential grocery categories consistently achieve higher customer retention and repeat purchases**, whereas discretionary and non-consumable categories generally experience lower reorder behaviour. This information can help the business prioritize inventory planning, promotional campaigns, and customer retention strategies for high-loyalty aisles while exploring initiatives to improve engagement in lower-performing categories.

Query 18 – Average Order Size by Day of the Week

Objective: To analyse how the average number of products purchased per order varies across different days of the week. This helps identify customer purchasing patterns and determine whether customers tend to place larger orders on specific days.

```
q18_avg_order_size_by_day = (  
    master_df  
    .groupby(["order_dow", "order_id"])  
    .agg(order_size=("product_id", "count"))  
    .reset_index()  
    .groupby("order_dow", as_index=False)  
    .agg(avg_order_size=("order_size", "mean"))  
    .sort_values("order_dow")  
)  
display(q18_avg_order_size_by_day)
```

	order_dow	avg_order_size
0	0	11.720820
1	1	10.292162
2	2	9.908470
3	3	9.667867
4	4	9.853087
5	5	10.198272
6	6	11.030755

Interpretation

- Customers place the **largest orders on Day 0**, with an average of **11.72 products per order**.
- Day 6** also records a relatively high average order size of **11.03 products**.
- The **smallest average order size** is observed on **Day 3**, with approximately **9.67 products per order**.
- Overall, the variation in average order size throughout the week is moderate, ranging from **9.67 to 11.72 products**, indicating relatively consistent purchasing behaviour.

Business Insight

Customers tend to purchase larger baskets on **Day 0 and Day 6**, suggesting these are major grocery shopping days. Retailers can leverage this pattern by ensuring adequate inventory, increasing staff availability, and promoting bundled offers or discounts during these high-volume shopping days to maximize sales and improve customer satisfaction.

Query 19 – Top 10 Users with the Highest Number of Orders

Objective: To identify the most active customers based on the total number of orders placed. This helps understand customer purchasing frequency and recognize highly engaged customers.

```
q19_top_users = (
    orders
    .groupby("user_id", as_index=False)
    .agg(total_orders=("order_id", "count"))
    .sort_values("total_orders", ascending=False)
    .head(10))
display(q19_top_users)
```

	user_id	total_orders
15895	15896	100
16289	16290	100
57837	57838	100
57850	57851	100
16297	16298	100
16312	16313	100
16339	16340	100
16356	16357	100
16363	16364	100
963	964	100

Interpretation

- The analysis identified the **top 10 customers** based on the total number of orders placed.
- Each of these customers has placed **100 orders**, making them the most active users in the dataset.
- The identical order count among the top customers suggests that **100 orders may represent the maximum order history available within the dataset** rather than an unusual purchasing pattern.
- These customers demonstrate consistent purchasing behaviour and can be considered highly engaged shoppers.

Business Insight

Highly active customers represent the business's most valuable customer segment. They are ideal candidates for loyalty programs, personalized offers, premium memberships, and targeted retention campaigns. Understanding the purchasing behavior of these customers can help improve customer lifetime value and strengthen long-term customer relationships.

Query 20 – Number of Products in Each Aisle and Department

Objective: To determine the distribution of products across each aisle within every department. This analysis helps understand the product assortment available in different categories and identifies aisles with the highest concentration of products.

```
q20_products_by_aisle_department = (
    products
    .merge(aisles, on="aisle_id")
    .merge(departments, on="department_id")
    .groupby(["department", "aisle"], as_index=False)
    .agg(total_products=("product_id", "count"))
    .sort_values(
        ["department", "total_products"],
        ascending=[True, False]))
display(q20_products_by_aisle_department)
```

	department	aisle	total_products
0	alcohol	beers coolers	384
1	alcohol	red wines	232
3	alcohol	spirits	192
4	alcohol	white wines	146
2	alcohol	specialty wines champagnes	95
...	...	...	...
128	snacks	fruit vegetable snacks	356
132	snacks	popcorn jerky	315
130	snacks	mint gum	168
129	snacks	ice cream toppings	83
133	snacks	trail mix snack mix	69

Total of 134 department–aisle combinations were identified.

Interpretation

- The analysis provides a detailed breakdown of product distribution across aisles within each department.
- Certain aisles contain significantly more products than others, indicating a broader product assortment within those categories.
- For example, within the **Alcohol** department, **Beers Coolers** contains the highest number of products, while in the **Snacks** department, **Fruit Vegetable Snacks** has the largest product assortment.

Business Insight

Understanding product distribution at both the department and aisle levels enables better category management, shelf space optimization, and inventory planning. Departments with large assortments should receive greater attention in merchandising strategies, while lower-volume aisles can be evaluated for assortment optimization or expansion based on customer demand.

Power BI Dashboard Development

After completing the Python-based analysis, the processed datasets were exported in CSV format to facilitate interactive dashboard development in Microsoft Power BI. The following datasets were exported:

- Aisles
- Departments
- Products
- Orders
- Order Products
- Master Dataset
- Top Reordered Products (Minimum 100 Orders)
- Most Reordered Product by Department

These datasets were imported into Power BI to create interactive dashboards and visualizations. The dashboards were designed to summarize key business metrics, customer purchasing behaviour, product performance, and department-wise analysis. The dashboard development process included:

- Exporting processed datasets from Python into CSV format.
- Importing the datasets into Microsoft Power BI.
- Establishing relationships between the imported tables where required.

- Creating calculated measures and KPI cards.
- Developing interactive visualizations and filters.
- Designing four dashboard pages to present the analytical findings in a structured and user-friendly format.

Power BI Dashboard

After completing the SQL and Python-based analysis, an interactive dashboard was developed in Microsoft Power BI to present the analytical findings in a clear and business-friendly format. To improve readability and avoid overcrowding, the dashboard was divided into four separate pages, with each page focusing on a specific aspect of the business. This approach enables decision-makers to quickly understand key metrics, customer behavior, product performance, and category-level insights.

The dashboard consists of the following four pages:

- Executive Business Analysis
- Customer Business Overview
- Product Performance Overview
- Department & Category Analysis

Executive Business Analysis Dashboard

Overview

The Executive Business Analysis Dashboard provides a high-level summary of the supermarket's overall performance. It combines key business metrics with product, customer, and order insights, allowing management to quickly understand the current business situation without exploring detailed reports.



Key Visualization 1: Product Distribution Across Departments

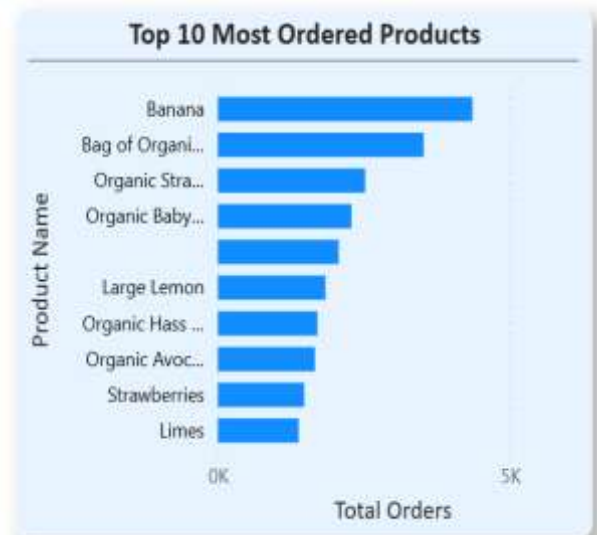
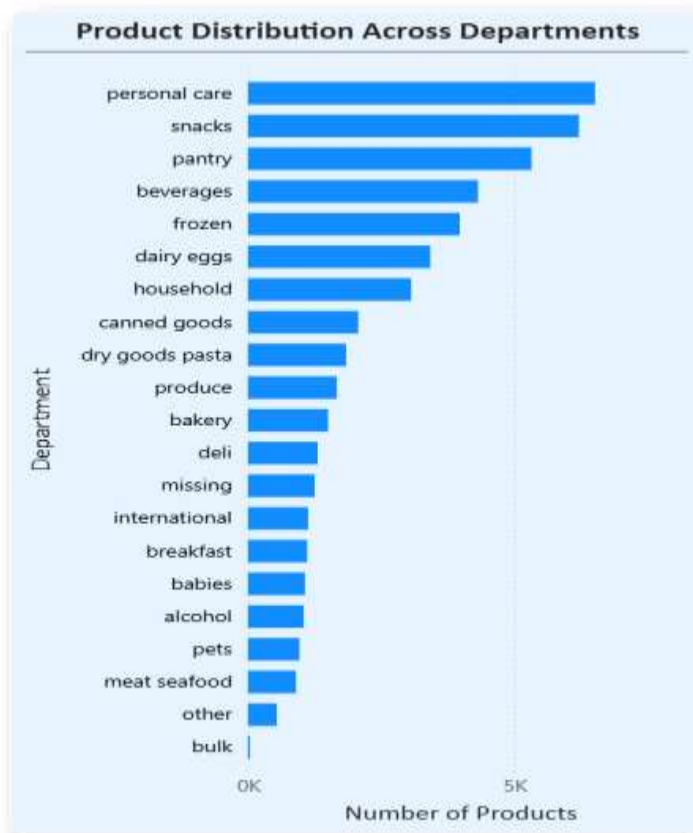
This chart displays the number of products available in each department. It helps identify departments with the largest product assortment and highlights how inventory is distributed across the business. Some Key Insights are:

- Personal Care contains the highest number of products.
- Snacks and Pantry also have a large product assortment.
- Bulk contains the fewest products, indicating a much smaller inventory compared to other departments.

Key Visualization 2: Top 10 Most Ordered Products

This visualization identifies the products that customers purchase most frequently. These products represent the highest demand items and play a significant role in overall sales performance. Some Key Insights are:

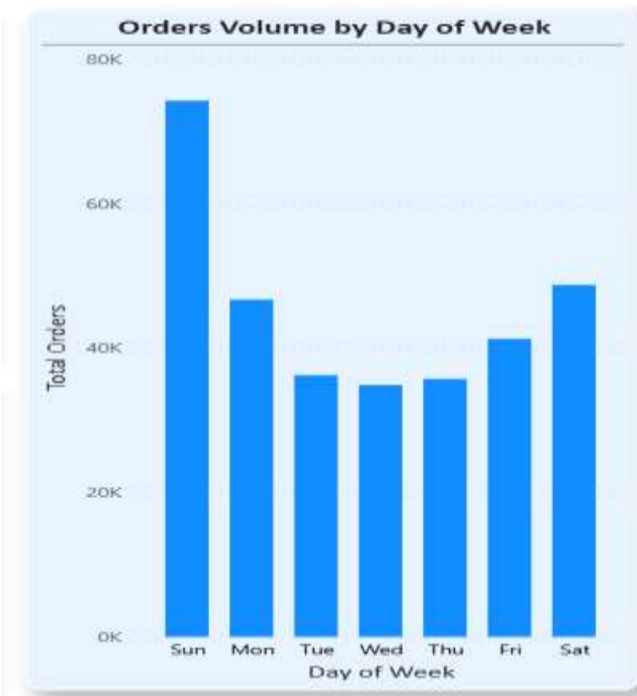
- Banana is the most frequently ordered product.
- Organic fruits and vegetables dominate the top-selling products.
- Fresh produce contributes significantly to customer purchasing behaviour.



Key Visualization 3: Orders Volume by Day of Week

This chart illustrates how customer order volume changes throughout the week. It helps identify peak shopping days and supports operational planning. Some Key Insights are:

- Customer activity is highest at the beginning of the week (based on the dataset's day mapping).
- Mid-week order volume is comparatively lower.
- Understanding weekly order patterns optimize staffing, inventory replenishment, and promotional plans.



Dashboard Summary

The Executive Business Analysis Dashboard provides a concise overview of the supermarket's overall performance by combining business KPIs, product distribution, customer activity, and order trends into a single page. It enables management to quickly identify high-performing product categories, understand customer purchasing patterns, and make informed decisions related to inventory management, promotions, and operational planning.

Key Findings

- 63,100 active customers analysed
- Peak ordering occurs between 9 AM–5 PM
- Sunday and Monday receive highest order volume
- Organic dairy and fresh produce dominate reorder behaviour
- Average basket size = 10.5 items
- 11,529 products reordered multiple times

Customer Business Analysis Dashboard

Overview

The Customer Business Analysis Dashboard focuses on understanding customer purchasing behavior and shopping patterns. It provides insights into customer distribution, ordering habits, purchase frequency, reorder behavior, and shopping trends across different departments and time periods. These insights help businesses optimize marketing strategies, inventory planning, and customer engagement initiatives.

Customer Business Analysis

Total Customers
63100



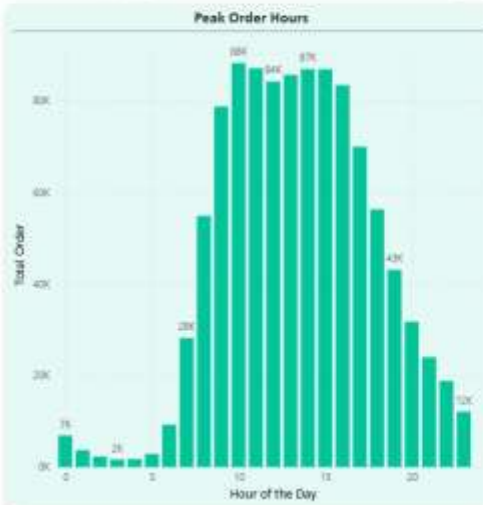
Average Products per Order
10.53



Average Days Between Orders
11.10



Average Reorder Rate
59.85%



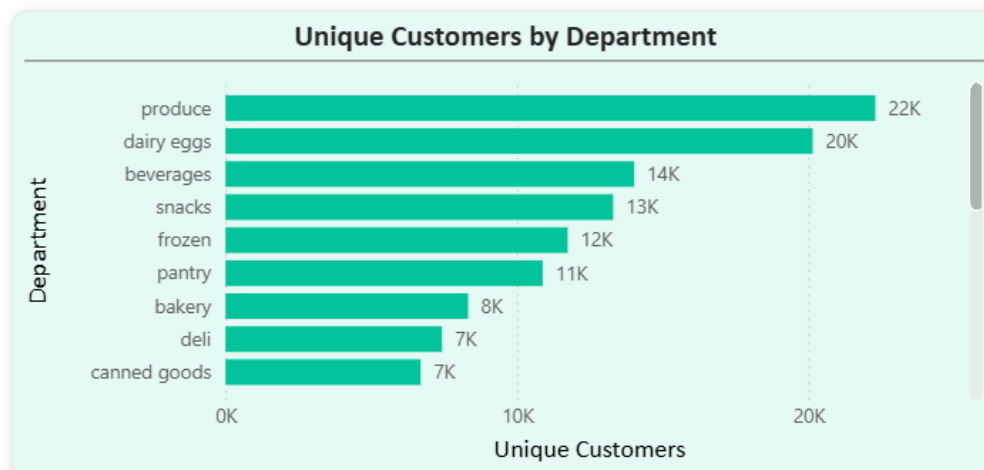
Filter By Department: All
☐ alcohol
☐ babies
☐ bakery



Key Visualization 1: Unique Customers by Department

This chart displays the number of unique customers purchasing products from each department. It helps identify departments that attract the largest customer base and highlights areas with the highest customer engagement. Some Key Insights are:

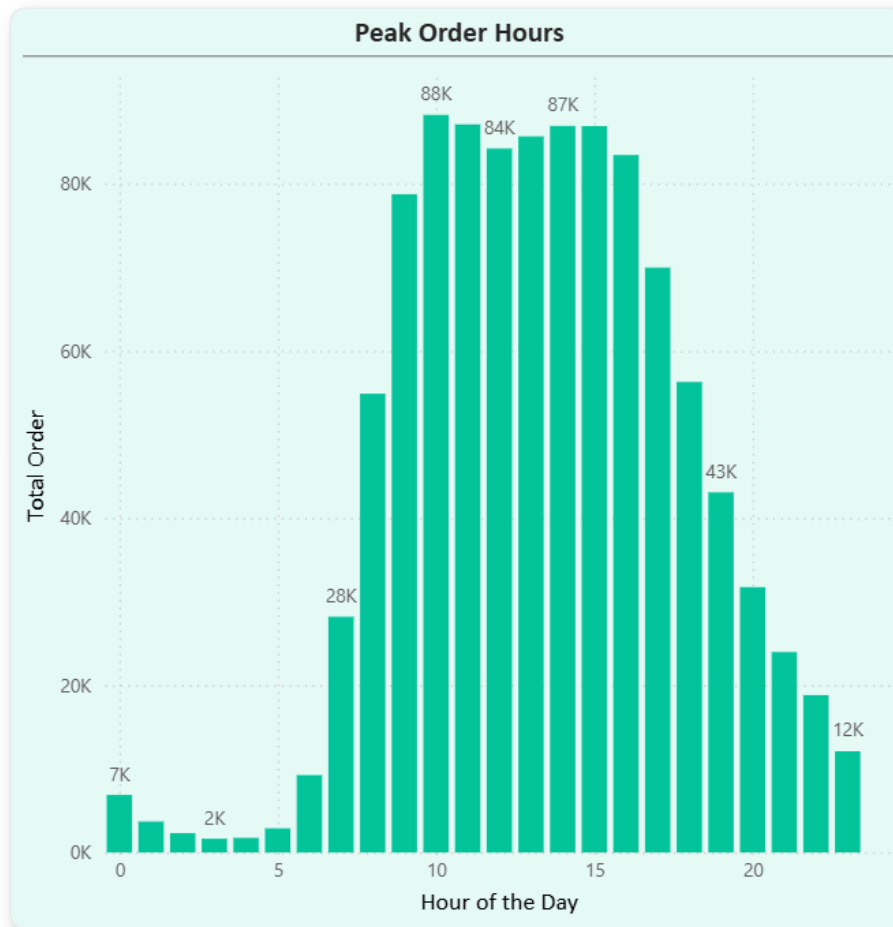
- Departments with higher customer reach present strong opportunities for promotions, cross-selling, and customer retention programs.
- Produce** attracts the highest number of unique customers.
- Dairy Eggs, Beverages, and Snacks** also have a large customer base.



Key Visualization 2: Peak Order Hours

This visualization shows the distribution of customer orders throughout the day, allowing the identification of peak shopping hours and periods of low customer activity. Some Key Insights are:

- Customer orders increase significantly during the morning hours.
- The highest order volume occurs between 10 AM and 3 PM, with the peak around 10 AM.
- Order activity declines steadily after the afternoon and reaches its lowest level during late-night and early-morning hours.



Key Visualization 3: Average Order Size by Day of Week

This chart illustrates the average number of products purchased per order on each day of the week, helping understand weekly shopping behavior. Some Key Insights are:

- Customers place the largest orders on **Sunday**, averaging approximately **11.7 products per order**.
- Average order size gradually decreases during the middle of the week.
- Order size begins increasing again toward the weekend, indicating that customers tend to make larger grocery purchases at the beginning and end of the week.



Dashboard Summary

The Customer Business Analysis Dashboard provides valuable insights into customer purchasing behavior by analyzing shopping frequency, department preferences, ordering patterns, and average basket size. The analysis reveals that Produce and Dairy departments attract the largest customer base, customer activity peaks between **10 AM and 3 PM**, and larger shopping baskets are typically observed on weekends. These findings can support decisions related to staffing, inventory replenishment, promotional timing, and customer relationship management.

Product Performance Analysis Dashboard

Overview

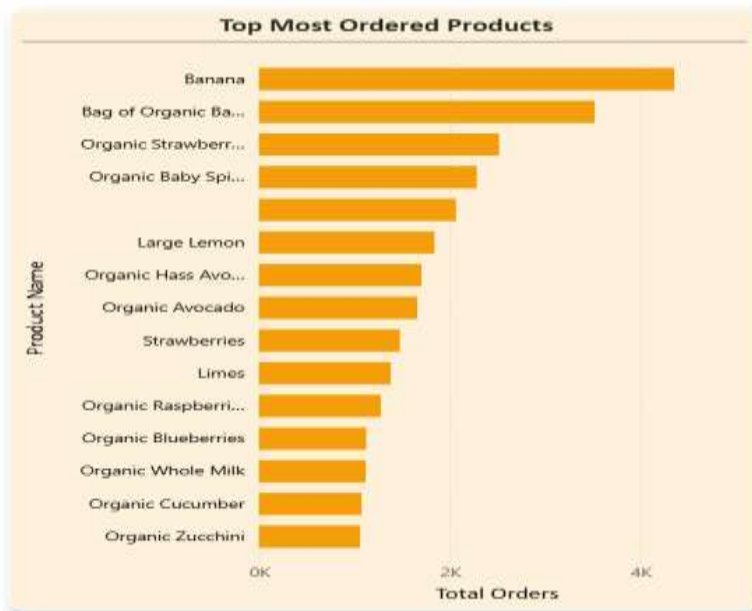
The Product Performance Analysis Dashboard focuses on evaluating product-level performance by analyzing customer purchasing patterns, reorder behavior, and product demand across different departments. It highlights the best-performing products, identifies products with high customer loyalty, and provides insights that support inventory management, merchandising, and product planning decisions.



Key Visualization 1: Top Most Ordered Products

This chart displays the products with the highest number of customer orders, helping identify the store's best-selling products and the primary drivers of overall sales. Some Key Insights are:

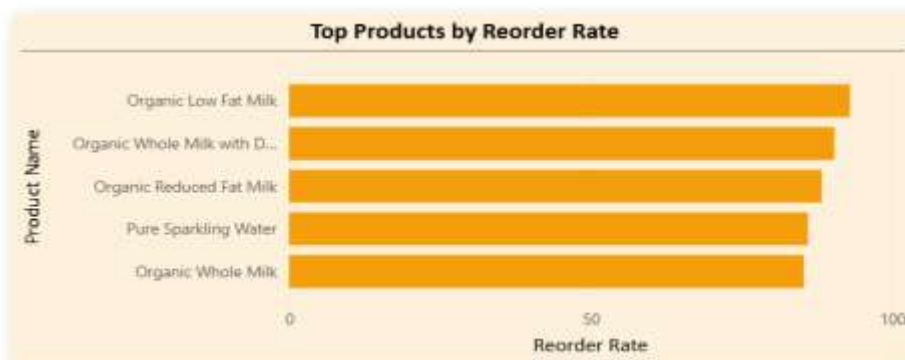
- **Banana** is the most ordered product, followed by **Bag of Organic Bananas** and **Organic Strawberries**.
- Fresh fruits and vegetables dominate the list of top-selling products.
- These products represent consistently high customer demand and require careful inventory planning to avoid stock shortages.



Key Visualization 2: Top Products by Reorder Rate

This visualization highlights products with the highest reorder rates after applying a minimum order threshold. This approach ensures that only products with sufficient purchase history are considered, making the analysis more reliable. Some Key Insights are:

- **Organic Low-Fat Milk** has the highest reorder rate, followed by **Organic Whole Milk with DHA Omega-3** and **Organic Reduced Fat Milk**.
- Organic dairy products demonstrate exceptionally strong customer loyalty.
- High reorder rates indicate products that customers repeatedly purchase and are ideal candidates for customer retention and loyalty campaigns.



Key Visualization 3: Most Reordered Product by Department

This tree map identifies the most reordered product within each department, providing a department-level view of customer preferences and product loyalty. Some Key Insights are:

- **Banana** is the most reordered product in the **Produce** department.
- **Organic Whole Milk** leads the **Dairy Eggs** department, while **Original Hummus** is the most reordered product in the **Deli** department.
- Every department has a distinct high-performing product that can be used as a flagship item for promotions and merchandising strategies.



Dashboard Summary

The Product Performance Analysis Dashboard provides a comprehensive evaluation of product demand and customer loyalty. The analysis reveals that fresh produce and organic dairy products consistently achieve the highest sales volume and reorder rates, indicating strong customer preference and repeat purchasing behaviour. These insights can support inventory optimization, promotional planning, assortment management, and customer retention strategies by focusing on high-performing products and departments.

Department & Category Analysis Dashboard

Overview

The Department & Category Analysis Dashboard provides a detailed view of product distribution and customer engagement across different departments and aisles. It helps evaluate category performance, identify high-performing departments, and understand how products are organized throughout the store. This dashboard supports strategic decisions related to inventory management, category optimization, and merchandising.

Department & Category Analysis

Total Departments

21



Total Aisles

134



Avg Products per Department

1.19K

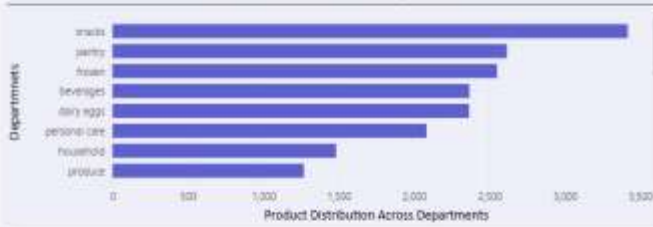


Largest Department

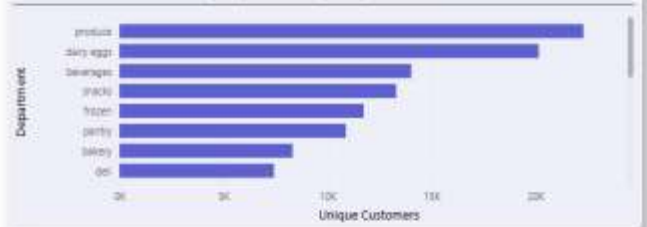
snacks



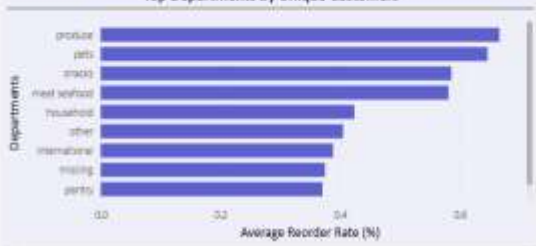
Product Distribution Across Departments



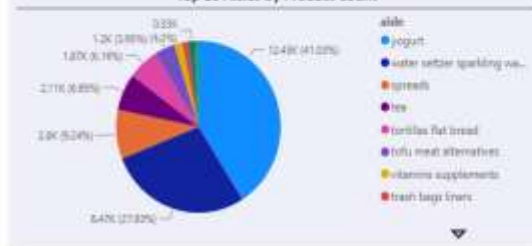
Top Departments by Unique Customers



Top Departments by Unique Customers



Top 10 Aisles by Product Count



Filter by Department: All

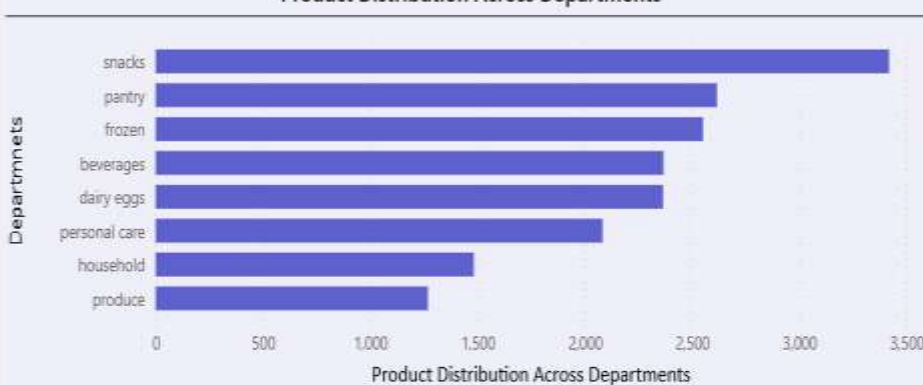


Key Visualization 1: Product Distribution Across Departments

This chart illustrates the number of products available in each department, providing an overview of how the overall product catalogue is distributed across business categories. Some Key Insights are:

- Snacks** contains the highest number of products among all departments.
- Pantry, Frozen, and Beverages** also maintain a large product assortment.
- Departments with larger inventories require more effective inventory management and shelf space planning.

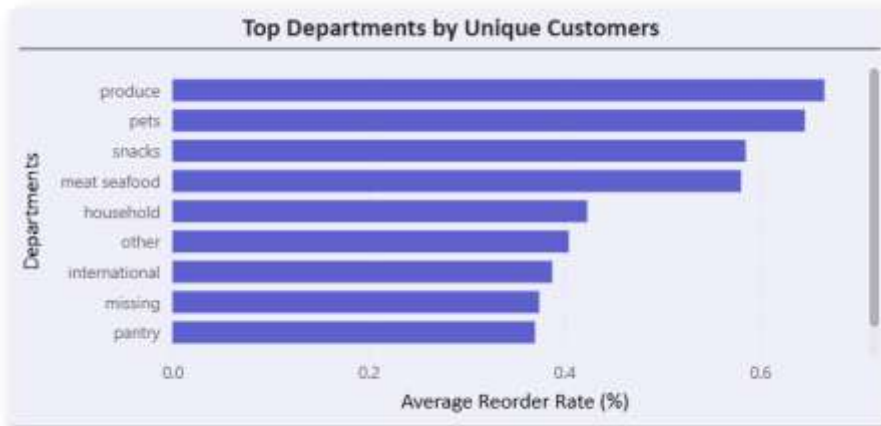
Product Distribution Across Departments



Key Visualization 2: Top Departments by Unique Customers

This visualization presents the departments that attract the highest number of unique customers, highlighting customer preferences across major product categories. Some Key Insights are:

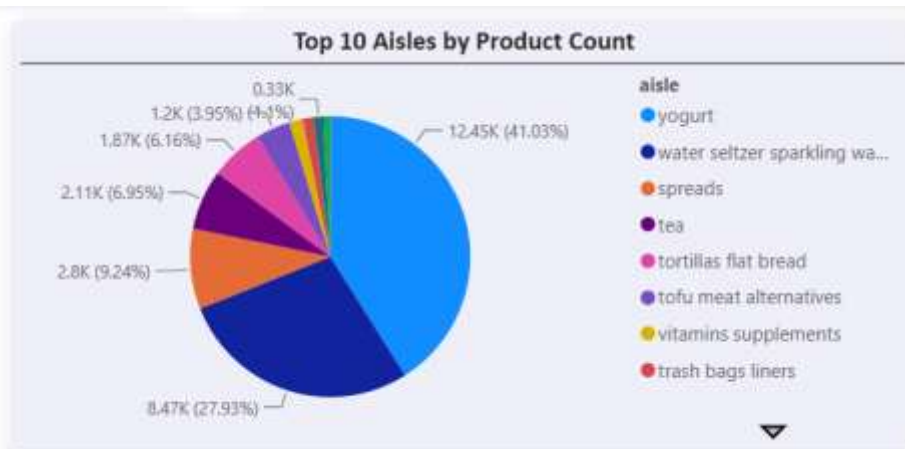
- **Produce** attracts the largest customer base, followed by **Dairy Eggs, Beverages, and Snacks**.
- These departments represent the store's primary customer traffic drivers.
- Focusing promotional campaigns and inventory investments on these departments can significantly improve customer satisfaction and sales performance.



Key Visualization 3: Top 10 Aisles by Product Count

This pie chart illustrates the distribution of products among the top 10 aisles with the highest product counts. It helps identify which aisles contribute the largest share of the overall product assortment and highlights categories with the greatest inventory concentration. Some Key Insights are:

- **Yogurt** represents the largest share of products among the top 10 aisles, accounting for approximately **41%** of the displayed product distribution.
- **Water Seltzer Sparkling Water** is the second-largest aisle, followed by **Spreads** and **Tea**, indicating that these aisles also maintain substantial product assortments.
- The remaining aisles each contribute a comparatively smaller proportion, suggesting that product availability is concentrated in a few major aisles.
- Understanding aisle-level product distribution helps optimize shelf allocation, category management, and inventory planning for high-volume product categories.



Dashboard Summary

The Department & Category Analysis Dashboard provides a comprehensive understanding of product distribution, customer engagement, and category performance across the supermarket. The analysis highlights that while **Snacks** offers the largest product assortment, **Produce** attracts the highest number of customers and demonstrates strong repeat purchasing behaviour. These insights support more effective category management, inventory allocation, merchandising strategies, and data-driven business decision-making.

Business Insights

A. Product Portfolio Insights

Product Catalogue and Distribution: The store features a broad and highly diversified product portfolio of 49,355 products across 21 departments and 134 aisles to cater to varied customer needs. However, product distribution is highly uneven; major categories like Personal Care (6,524), Snacks (6,221), and Pantry (5,331) hold the largest shares, whereas the Bulk department is the smallest with only 38 products. *Suggested Business Impact:*

- Inventory management efforts should focus more on departments with large assortments.
- Smaller departments may require assortment optimization.

Aisle Variety: Certain aisles exhibit extremely large product variety, led by Missing (1,254), Candy Chocolate (1,235), Ice Cream (1,084), and Vitamins & Supplements (1,030). *Suggested Business Impact:*

- Large aisles require better shelf planning.
- Higher assortment may increase customer choice but also operational complexity.

B. Customer Base Insights

Customer Base: The store maintains a large active customer base, with the dataset capturing 63,100 unique customers. *Suggested Business Impact:*

- Large enough customer base for segmentation and personalized marketing.

Purchase frequency: Customers purchase regularly, with the average time between transactions typically ranging from 10 to 25 days. *Suggested Business Impact:*

- Customers exhibit recurring purchasing behaviour.
- Good opportunity for reminder campaigns and subscription models.

C. Purchasing Behaviour

Customers Prefer Shopping During Midday: Customer orders peak during the midday hours of 10 AM, 11 AM, 2 PM, and 3 PM, while the lowest transaction volumes occur during the early morning between 2 AM and 5 AM. *Suggested Business Impact:*

- Schedule staff during peak hours.
- Launch promotional notifications before peak ordering windows.

Weekends Are Busier: Order volume peaks during the weekend on Saturday and Sunday, while Wednesday experiences the lowest traffic. *Suggested Business Impact (Interpret according to the dataset's day mapping.):*

- Increase inventory and staffing during busy days.

Customers Buy Products Per Order: On average, customers purchase 10.51 products per order. *Suggested Business Impact:*

- Customers generally purchase medium-sized baskets.
- Cross-selling opportunities exist.

Order Sizes Vary Widely: Order sizes vary widely, ranging from 1 to 74 products, though most customers purchase relatively small-to-medium baskets. *Suggested Business Impact:*

- Large orders represent valuable customers.
- Bundle offers may encourage larger baskets.

D. Product Loyalty

Organic Products Show Strong Customer Loyalty: Organic dairy products demonstrate exceptionally strong customer loyalty when applying a minimum threshold of 100 orders. For instance, Organic Low-fat Milk, Organic Whole Milk, Organic Whole Milk DHA, and Organic Reduced Fat Milk all maintain reorder rates exceeding 85%. *Suggested Business Impact:*

- Organic products drive repeat purchases.

Bananas Dominate the Store: Bananas dominate the store's inventory performance, ranking repeatedly as both the single highest-ordered product overall and the highest-reordered item within the Produce department. *Suggested Business Impact:*

- Bananas are a core demand driver.
- Inventory shortages could directly affect customer satisfaction.

Different Departments Have Different Loyalty Drivers: Each department is anchored by its own specific hero product that drives customer loyalty. Notable examples include Bananas in Produce, Organic Whole Milk in Dairy, Original Hummus in Deli, and Sauvignon Blanc in Alcohol. *Suggested Business Impact:*

- Department-specific promotions should focus on these products.

E. Customer Reach

Produce Reaches the Largest Number of Customers

The Produce department reaches the largest number of customers, followed closely by Dairy Eggs, Beverages, and Snacks as the categories with the highest unique customer counts. *Suggested Business Impact:*

- These departments attract the broadest customer base.
- They deserve greater promotional investment.

F. Product Demand

Fresh Produce Dominates Orders: Fresh produce dominates customer orders, with top-ordered products consisting of bananas, organic bananas, organic strawberries, organic baby spinach, avocados, and lemons. *Suggested Business Impact:*

- Fresh produce drives customer traffic.
- Demand forecasting is especially important for perishables.

G. Reordering Behaviour

High Reorder Rates Need Context: While several products initially showed a 100% reorder rate, these figures were skewed by low purchase volumes; applying a minimum threshold of 100 orders established a much more reliable metric. *Suggested Business Impact:*

- Business decisions should not rely solely on percentages; sufficient sample size is essential for reliable conclusions.

Key Variables from Cross Analysis

- Departments with the largest product assortment are **not necessarily** the ones with the highest reorder rates. **Meaning:** A wider assortment does not automatically create stronger customer loyalty.
- Produce has comparatively fewer products than Personal Care, yet it serves the **largest number of unique customers**. **Meaning:** Produce items are everyday essentials with broad customer appeal.
- Organic dairy products consistently appear among the highest reorder-rate products.
- **Meaning:** Customers who buy organic dairy are highly loyal.
- Peak shopping hours (10 AM–3 PM) align with medium basket sizes (~10 products), indicating customers are likely doing planned grocery shopping rather than impulse purchases.
- Fresh produce dominates both **most ordered products** and **most reordered products**, reinforcing its importance in overall store performance.
- Some departments (e.g., Bulk, Pets) have relatively few products and fewer customers, suggesting they contribute less to overall engagement than major grocery categories.
- High-performing departments (Produce, Dairy Eggs, Beverages, Snacks) combine **large customer reach** with **strong repeat purchasing**, making them strategic focus areas for promotions and inventory planning.

Business Recommendations

A. Prioritize Inventory for High-Demand Products

Observation: Products like Banana, Organic Bananas, Organic Strawberries, Organic Baby Spinach, and Organic Whole Milk consistently rank among the most ordered and reordered items.

Recommendation

- Maintain higher safety stock for these products.
- Use demand forecasting to prevent stockouts.
- Monitor inventory daily for these fast-moving products.

Expected Benefit

- Higher product availability.
- Increased customer satisfaction.
- Reduced lost sales.

B. Strengthen Organic Product Portfolio

Observation: Organic dairy and produce products have exceptionally high reorder rates.

Recommendation

- Expand organic product offerings.
- Promote organic bundles.
- Feature organic products in marketing campaigns.

Expected Benefit

- Improved customer retention.
- Higher repeat purchases.
- Increased revenue from loyal customers.

C. Optimize Staffing During Peak Hours

Observation: Most orders occur between **10 AM and 3 PM**.

Recommendation

- Increase staffing during peak shopping hours.
- Schedule replenishment before peak demand.
- Allocate more delivery resources during these hours.

Expected Benefit

- Faster service.
- Reduced customer waiting time.
- Better operational efficiency.

D. Increase Weekend Readiness

Observation: Order volume is highest on **Day 0 and Day 1**.

Recommendation

- Increase inventory before weekends.
- Schedule additional staff.
- Run weekend promotional campaigns.

Expected Benefit

- Better customer experience.
- Increased sales.
- Reduced stock shortages.

E. Focus Promotions on High-Customer Departments

Observation: Produce, Dairy Eggs, Beverages, and Snacks attract the highest number of unique customers.

Recommendation

- Launch targeted promotions in these departments.
- Place featured products at store entrances.
- Offer bundle discounts across these departments.

Expected Benefit

- Increased customer engagement.
- Higher average basket value.
- More cross-selling opportunities.

F. Improve Low-Performing Departments

Observation: Departments like **Bulk, Pets, Alcohol, and Other** have fewer products and fewer customers.

Recommendation

- Review product assortment.
- Remove consistently underperforming products.
- Test new products based on customer demand.

Expected Benefit

- Better inventory utilization.
- Lower holding costs.
- Improved department performance.

G. Increase Average Basket Size

Observation: Customers purchase an average of **10.5 products per order**.

Recommendation

- Introduce "Frequently Bought Together" recommendations.
- Offer "Buy More, Save More" promotions.
- Create meal kits and bundled product offers.

Expected Benefit

- Higher average order value.
- Increased revenue per customer.

H. Build Customer Loyalty Programs

Observation: Many products have high reorder rates, indicating strong repeat purchasing behavior.

Recommendation

- Introduce personalized loyalty rewards.
- Offer repeat-purchase discounts.
- Send reorder reminders based on purchase frequency.

Expected Benefit

- Higher customer retention.
- Increased lifetime customer value.
- More recurring sales.

I. Improve Shelf Space Allocation

Observation: Personal Care, Snacks, and Pantry contain the largest number of products.

Recommendation

- Allocate shelf space based on product demand rather than only product count.
- Regularly review category performance to optimize layouts.

Expected Benefit

- Better product visibility.
- Improved shopping experience.
- Increased sales from high-demand categories.

J. Use Customer Purchase Patterns for Marketing

Observation: Customers exhibit consistent purchasing cycles and predictable shopping behaviour.

Recommendation

- Schedule email campaigns and app notifications before peak shopping periods.
- Personalize offers using previous purchase history.
- Recommend complementary products during checkout.

Expected Benefit

- Higher campaign conversion rates.
- Increased repeat purchases.
- Better customer engagement.

K. Develop Department-Specific Marketing Strategies

Observation: Each department has a distinct top-performing product (e.g., Banana in Produce, Organic Whole Milk in Dairy).

Recommendation

- Design promotions around each department's best-selling product.
- Use these products as "traffic drivers" to encourage additional purchases within the department.

Expected Benefit

- Increased department-level sales.
- Higher cross-selling opportunities.

L. Base Business Decisions on Reliable Metrics

Observation: Products with very few purchases can show misleadingly high reorder rates.

Recommendation

- Apply a minimum order threshold (e.g., 100 orders) before evaluating product loyalty.
- Combine reorder rate with total order volume when identifying high-performing products.

Expected Benefit

- More reliable decision-making.
- Reduced risk of acting on statistically weak patterns.

Conclusion & Future Scope

This analysis converts a relational grocery-order dataset into a validated, cross-checked picture of what's bought, how often, by whom, and when — anchored by SQL-level integrity checks, complete pandas answers to all 20 brief questions, and four Power BI dashboards for stakeholder self-service. The single most decision-ready finding is the **convergent evidence across four independent metrics** that a small set of perishable staples anchor customer engagement, while catalog breadth sits elsewhere — this insight alone should drive the next round of assortment and retention-marketing planning.

Future Scope

- **Validate the extraction cap** — confirm whether the ~1,048,575-row tables represent a capped extract and, if so, re-run this analysis against the full order history.
- **Enrich with pricing/margin and marketing-channel data** so reorder-rate insights can be turned into ROI-ranked recommendations rather than volume-ranked ones.
- **Extend to a rolling/cohort view** — e.g. reorder cadence by customer tenure — once more than a 10-user sample is available for Q6-style analysis.
- **Resolve the "missing" category and the small Power BI/Python metric discrepancies** as a formal data-quality remediation backlog item.
- **Move from a point-in-time CSV export to a refreshable Power BI model** so the dashboards stay current as new orders arrive, rather than reflecting a single export snapshot.